

MANUEL PAEZ

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Education

Columbia University	New York, NY
Master of Science, Computer Science , GPA: 3.8/4.0	Jan. 2025 - Expected May 2026
Bachelor of Arts, Mathematics	Sept. 2023 - Expected May 2026
Bachelor of Arts, Computer Science	Sept. 2019 - May 2023
Relevant Coursework: Advanced Probability Theory, Algorithms for Massive Data, Analysis and Probability, Machine Learning Theory, Log-Concave Probability, Unsupervised Learning, Theoretical Foundations of LLMs.	

Work and Research Experience

Flatiron Institute - Simons Foundation	New York, NY
Visiting Scholar Fellow , Neural Circuits and Algorithms Group	Nov. 2024 - Aug. 2025
Developing algorithmic and diffusion-based models for nonlinear neuronal dynamical systems, with emphasis on theoretical guarantees and numerical stability.	

Columbia University	New York, NY
Course Assistant , Department of Computer Science	Sept. 2024 - Present
Taught Recitations, held office hours, and graded assignments for COMS 4995: Randomized Algorithms and COMS 6998: Algorithms for Massive Data.	

Flatiron Institute - Simons Foundation	New York, NY
Research Intern , Neural Circuits and Algorithms Group	May 2024 - Aug. 2024
- Redesigned CaImAn's large-scale calcium imaging data analysis tool from Tensorflow using PyTorch and Keras. - Reintegrated an automated and scalable analysis pipeline for voltage imaging datasets with a PyTorch backend. - Utilized Flatiron Institute's HPC cluster to train, test, and execute PyTorch-implemented supervised machine learning models on 10+ TB calcium imaging datasets. Code predominantly in Python. Coordinated with Janelia scientists.	

Flatiron Institute - Simons Foundation	New York, NY
Research Intern , Neural Circuits and Algorithms Group	Sept. 2022 - Aug 2023
- Developed and evaluated a supervised IsoRSUNet architecture to improve neuron-boundary segmentation in assembled connectome maps. Conducted ablation experiments to analyze architectural stability and generalization. - Leveraged Flatiron Institute's HPC cluster to automate segmentation processes. Code predominantly in Python.	

Awards

ClimateScience Olympiad 2024, Semi-finalist	Aug. 2024
MIT IQuHACK Hackathon 2023; Covalent x IBM Challenge - 1st place	Jan 2023
Simon Foundation Global Brain SURF Fellowship	Aug 2022

Projects and Extracurricular Activities

Unsupervised Learning , "Uncovering User Information from Obfuscated K-Clustering"	Oct. 2024 - Dec. 2024
Collaborated on a theoretical and empirical framework for information reconstruction attacks on K-clustering.	
Undergraduate Project in CS , "Quantum Algorithm for Detecting Gravitational Waves"	Nov. 2022 - May 2023
Executed in Python an innovative quantum advantage pattern recognition method on LIGO gravitational-wave data.	
Columbia Undergraduate Quantum Computing Club	Nov. 2022 - Jan. 2023
Co-founder and former co-head; led the hackathon team to 1st place at MIT IQuHACK 2023 IBM x Covalent Event.	

Technical Skills

Languages and Frameworks: Python, C/C++, Java, MATLAB, R, PyTorch, Tensorflow, Jupyter, Keras, HPC, Qiskit